

1. Applied Research > General > Model Evaluation
2. In my thesis on diffusion features for pose estimation, one of my ideas was to use a depth control net as proxy for geometric understanding since the branch is trained with depth on additional encoder branch. However after initial experimentation and discussion with my advisor we realized that the training objective and architecture of the control net merely refines the diffusion RGB features in the same latent space as original features and depth information added which is necessary for correspondence task is marginal.
3. During my time in AbInBev testing whether the modification in architectures actually learnt the intended effects was a huge problem since we rarely ever had clean labeled data and often had to rely on domain experts' knowledge to validate the model behaviour. I took initiative to create a parametrized synthetic data generator where we can exactly simulate the varying levels of any effect on fake data and measure if model architecture is able to capture that. The generator learnt distributions from ground truth data and allowed us to modify the parameters and add additional components on top of existing parameters. As an fringe benefit we were as able to integrate this into our CI/CD pipeline to ensure the model behavior does not break with any changes pushed by developers
4. When it comes to weather prediction there are a few cases that come to my mind where accuracy hacking might be detrimental
 1. If we view this as an parallel to anomaly detection...the model should be useful for predicting bad weather which might be uncommon but accuracy can be hacked by predicting more of normal weather however this is gross implication of a weather model
 2. Since weather prediction is a high uncertainty problem, confidence of the model is also important and accuracy hacking without confidence calibration can lead to less useful results
5. My views on AI for coding is a bit regressive vs how most people use it. I observed the way coding agents do abstraction and classes in codebase is often without a regard for future scaling. I since then have switched to a workflow involving design the classes and method with a lot intervention from my end instead of giving the agent full control and let it run inside a testing loop. However, I do think my view will change soon given the rate of progress in how the frontier models are performing
6. I mostly follow alphaxiv for keeping track of latest research <https://x.com/askalphaxiv?s=20> but I also love Yan LeCun's takes on his twitter handle.